# MSG No. 110.250 $I_c4_1cd$ [ Type IV, tetragonal ]

Table 1: Wyckoff site: `8a`, site symmetry: `2m'm'.`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,4,21,22]` |
| 2 | $\left[0,\ \frac{1}{2},\ z+\frac{1}{4}\right]$ | `[2,3,31,32]` |
| 3 | $\left[0,\ 0,\ z+\frac{1}{2}\right]$ | `[5,6,17,20]` |
| 4 | $\left[\frac{1}{2},\ 0,\ z+\frac{1}{4}\right]$ | `[7,8,26,27]` |
| 5 | $\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[9,12,29,30]` |
| 6 | $\left[\frac{1}{2},\ 0,\ z+\frac{3}{4}\right]$ | `[10,11,23,24]` |
| 7 | $\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$ | `[13,14,25,28]` |
| 8 | $\left[0,\ \frac{1}{2},\ z+\frac{3}{4}\right]$ | `[15,16,18,19]` |

Table 2: Wyckoff site: `16b`, site symmetry: `2'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$ | `[1,28]` |
| 2 | $\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{4}\right]$ | `[2,27]` |
| 3 | $\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{4}\right]$ | `[3,26]` |
| 4 | $\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$ | `[4,25]` |
| 5 | $\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$ | `[5,30]` |
| 6 | $\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$ | `[6,29]` |
| 7 | $\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{4}\right]$ | `[7,32]` |
| 8 | $\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{4}\right]$ | `[8,31]` |
| 9 | $\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$ | `[9,20]` |
| 10 | $\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{3}{4}\right]$ | `[10,19]` |
| 11 | $\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{3}{4}\right]$ | `[11,18]` |
| 12 | $\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$ | `[12,17]` |
| 13 | $\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$ | `[13,22]` |
| 14 | $\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$ | `[14,21]` |
| 15 | $\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{3}{4}\right]$ | `[15,24]` |
| 16 | $\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{3}{4}\right]$ | `[16,23]` |

Table 3: Wyckoff site: `16c`, site symmetry: `.m'.`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ y,\ z]$ | `[1,21]` |
| 2 | $\left[-y,\ \frac{1}{2},\ z+\frac{1}{4}\right]$ | `[2,31]` |
| 3 | $\left[y,\ \frac{1}{2},\ z+\frac{1}{4}\right]$ | `[3,32]` |
| 4 | $[0,\ -y,\ z]$ | `[4,22]` |
| 5 | $\left[0,\ y,\ z+\frac{1}{2}\right]$ | `[5,17]` |
| 6 | $\left[0,\ -y,\ z+\frac{1}{2}\right]$ | `[6,20]` |
| 7 | $\left[\frac{1}{2}-y,\ 0,\ z+\frac{1}{4}\right]$ | `[7,26]` |
| 8 | $\left[y+\frac{1}{2},\ 0,\ z+\frac{1}{4}\right]$ | `[8,27]` |
| 9 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[9,29]` |
| 10 | $\left[\frac{1}{2}-y,\ 0,\ z+\frac{3}{4}\right]$ | `[10,23]` |

*continued ...*

Table 3

| No. | position | mapping |
|---|---|---|
| 11 | $\left[y+\frac{1}{2},\ 0,\ z+\frac{3}{4}\right]$ | [11,24] |
| 12 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [12,30] |
| 13 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$ | [13,25] |
| 14 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$ | [14,28] |
| 15 | $\left[-y,\ \frac{1}{2},\ z+\frac{3}{4}\right]$ | [15,18] |
| 16 | $\left[y,\ \frac{1}{2},\ z+\frac{3}{4}\right]$ | [16,19] |

Table 4: Wyckoff site: 32d, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1] |
| 2 | $\left[-y,\ x+\frac{1}{2},\ z+\frac{1}{4}\right]$ | [2] |
| 3 | $\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{4}\right]$ | [3] |
| 4 | $[-x,\ -y,\ z]$ | [4] |
| 5 | $\left[-x,\ y,\ z+\frac{1}{2}\right]$ | [5] |
| 6 | $\left[x,\ -y,\ z+\frac{1}{2}\right]$ | [6] |
| 7 | $\left[\frac{1}{2}-y,\ -x,\ z+\frac{1}{4}\right]$ | [7] |
| 8 | $\left[y+\frac{1}{2},\ x,\ z+\frac{1}{4}\right]$ | [8] |
| 9 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [9] |
| 10 | $\left[\frac{1}{2}-y,\ x,\ z+\frac{3}{4}\right]$ | [10] |
| 11 | $\left[y+\frac{1}{2},\ -x,\ z+\frac{3}{4}\right]$ | [11] |
| 12 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [12] |
| 13 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$ | [13] |
| 14 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$ | [14] |
| 15 | $\left[-y,\ \frac{1}{2}-x,\ z+\frac{3}{4}\right]$ | [15] |
| 16 | $\left[y,\ x+\frac{1}{2},\ z+\frac{3}{4}\right]$ | [16] |
| 17 | $\left[x,\ y,\ z+\frac{1}{2}\right]$ | [17] |
| 18 | $\left[-y,\ x+\frac{1}{2},\ z+\frac{3}{4}\right]$ | [18] |
| 19 | $\left[y,\ \frac{1}{2}-x,\ z+\frac{3}{4}\right]$ | [19] |
| 20 | $\left[-x,\ -y,\ z+\frac{1}{2}\right]$ | [20] |
| 21 | $[-x,\ y,\ z]$ | [21] |
| 22 | $[x,\ -y,\ z]$ | [22] |
| 23 | $\left[\frac{1}{2}-y,\ -x,\ z+\frac{3}{4}\right]$ | [23] |
| 24 | $\left[y+\frac{1}{2},\ x,\ z+\frac{3}{4}\right]$ | [24] |
| 25 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$ | [25] |
| 26 | $\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{4}\right]$ | [26] |
| 27 | $\left[y+\frac{1}{2},\ -x,\ z+\frac{1}{4}\right]$ | [27] |
| 28 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$ | [28] |
| 29 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [29] |
| 30 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [30] |
| 31 | $\left[-y,\ \frac{1}{2}-x,\ z+\frac{1}{4}\right]$ | [31] |
| 32 | $\left[y,\ x+\frac{1}{2},\ z+\frac{1}{4}\right]$ | [32] |